Wkst A Instructions (updated September 2025)

All problems must have work to receive any points. All equations are to be solved on [0, 2π]. Level 3 and Level 4 equations must also include the minimum number of general solutions.

Separate problems into Stage 1, Stage 2/Stage 3. Do not renumber your problems; use the actual number. Papers not organized correctly will not be scored.

Questions will be scored on accuracy of work. You will receive full points / partial points / no points for each question. Minor errors will receive half points; major errors (like violating the laws of algebra or trig) will not receive any points.

Stage 1:

You must complete one problem from each type at each level. (15 problems total). The Wkst A Problem Picker (http://oconnorclass.com/aat/wksta.php) will tell you which problem is required for you to do in each box. Print the web page that is generated, IN PORTRAIT MODE, and scaled to fit on one page, to be the Cover Sheet for the Wkst A packet you will turn in. Correct work shown in an organized manner with proper mathematical notation will earn 61 points. These problems must be listed in numerical order. Scoring – L1: 1/0/0, L2: 3/2/0, L3: 5/3/0, and L4: 8/4/0

Stage 2:

Pick and complete additional problems to earn up to 100 points. Work must be correct and organized with proper mathematical notation. List the problems in numerical order. Scoring (same as Stage 1) – L1: 1/0/0, L2: 3/2/0, L3: 5/3/0, and L4: 8/4/0

Stage 3 – Bonus:

You may complete additional problems to A) get your total up to 100 points if your Stage 1 and Stage 2 problems do not yet total 100 points and/or B) earn more than 100 points. After you reach 100 possible points, Level 1 and Level 2 questions will be worth 1 Bonus point, Level 3 questions are worth 2 Bonus points, and Level 4 questions are worth 4 bonus points. The entire problem, work and answers, must be correct to earn the bonus point(s). Scoring – L1: 1/0/0, L2: 1/0/0, L3: 2/0/0, and L4: 4/0/0

In Stage 2 and Stage 3, list the problems in numerical order.

Your max grade, with bonus, can be up to 130 points. Worksheet A will count as a single formative. Worksheet A MUST BE TURNED IN ON TIME to qualify for bonus points. If you are absent on your test day for an EXCUSED reason (will be verified through Attendance records), it must be turned in on the very next day you are at school, even if your class does not meet that day (i.e. bring it to your teacher's room and turn it in).

	Level 1: Beginning 1 pts each	Level 2: Approaching 3 pts each	Level 3: Proficient 5 pts each	Level 4: Distinguished 8 pts each
Isolate	1. $2\cos x - 1 = 0$ 2. $\sqrt{2}\sin x - 1 = 0$ 3. $\tan x - \sqrt{3} = 0$ 4. $2\sin x + 1 = 0$ 5. $\tan x - 1 = 0$ 6. $3\sqrt{3}\tan x = 3$	7. $\sqrt{3}\sec x + 2 = 0$ 8. $\cot x - \sqrt{3} = 0$ 9. $\sin x - \sqrt{3} = -\sin x$ 10. $4\sec x - 2 = 2\sec x$ 11. $\frac{1}{2}\sec x - 1 = 0$	12. $2\sin x - \csc x = 0$ 13. $\cos x + \sin x \tan x = 2$ 14. $2\left(\frac{\sin x}{\tan x}\right) - 1 = 0$ 15. $\frac{\csc(-x)}{\sec(-x)} = 1$ 16. $\cot\left(\frac{\pi}{2} - x\right)\csc x - 2 = 0$	17. $\frac{\cos x \cot x}{1 - \sin x} = 3$ 18. $\frac{1+\sin x}{\cos x} + \frac{\cos x}{1+\sin x} = 4$ 19. $\frac{\cot^2 x}{1+\csc x} = 1$ 20. $\cos 2x + 2\sin x - 4\sin^3 x = 0$ 21. $2\sin x + \csc x = 0$
By Square Roots	22. $4\cos^2 x - 1 = 0$ 23. $\tan^2 x - 1 = 0$ 24. $4\cos^2 x - 3 = 0$	25. $3\csc^2 x - 4 = 0$ 26. $3\cot^2 x - 1 = 0$ 27. $\csc^2 x - 2 = 0$ 28. $4\tan^2 x - 1 = \tan^2 x$	29. $\sin^2 x = 3\cos^2 x$ 30. $2\sec^2 x + \tan^2 x - 3 = 0$ 31. $\sec^2 x + 2\tan^2 x = 4$ 32. $4\left(\frac{\cot x}{\csc x}\right)^2 - 3 = 0$ 33. $\sin^2\left(\frac{\pi}{2} - x\right) + \sin^2 x + \sec^2 x = 3$	
By Factoring	34. $\cos x(\cos x - 1) = 0$ 35. $\left(\sqrt{3}\tan x - 1\right)\left(\tan x - \sqrt{3}\right) = 0$ 36. $\sin x(6\sin x + 3) = 0$	37. $3\tan^3 x = \tan x$ 38. $\sec^2 x - \sec x = 2$ 38b. $\sec x \csc x = 2\csc x$ 39. $2\cos^2 x + \cos x - 1 = 0$ 40. $2\sin^2 x + 3\sin x + 1 = 0$ 41. $2\sin^2 x + 5\sin x = 3$	42. $2\sin^2 x = 2 + \cos x$ 43. $\sin^2 x + \cos x + 1 = 0$ 44. $\sec^2 x + \tan x = 3$ 45. $2\cos x + \sin 2x = 0$ 46. $\sin 2x + \cos x = 0$ 47. $\sin 2x \sin x = \cos x$ 48. $\cos 2x - \cos x = 0$ 49. $\tan x + 1 = \sqrt{3} + \sqrt{3}\cot x$	

	Level 1: Beginning 1 pts each	Level 2: Approaching 3 pts each	Level 3: Proficient 5 pts each	Level 4: Distinguished 8 pts each
By Squaring Both Sides			50. $\sin x - \cos x = 1$ 51. $\sin x + \cos x = -1$ 52. $\tan x + \sec x = 1$	
Multiple Angles		53. $\sin 2x = -\frac{1}{\sqrt{2}}$ 54. $\cos 3x = 0$ 55. $\sqrt{3}\tan 3x = 0$	56. $2\sin 3x \cos 2x + 2\sin 3x - \cos 2x - 1 = 0$ 57. $3\csc^2 5x - 4 = 0$ 58. $\sin x \cos x = \frac{\sqrt{2}}{4}$	
Higher Order Identities			59. $\cos\left(x + \frac{\pi}{6}\right) - \cos\left(x - \frac{\pi}{6}\right) = 1$ 60. $\text{Sin}\left(x + \frac{\pi}{3}\right) + \sin\left(x - \frac{\pi}{3}\right) = 1$ 61. $\sin 4x - \sin 2x = 0$ 62. $\sin 6x + \sin 2x = 0$	63. $\cos 4x - 7\cos 2x = 8$ 64. $\sin^2 3x - \sin^2 x = 0$ 65. $\frac{\cos 2x}{\sin 3x - \sin x} = 0$ 66. $2\sin\left(x + \frac{\pi}{2}\right) + 3\tan(\pi - x) = 0$ 67. $\sin 3x \cos 2x + \cos 3x \sin 2x = 1$

Remember: Worksheet A MUST BE TURNED IN ON TIME to qualify for bonus points. If you are absent on your test day for an EXCUSED reason (will be verified through Attendance records), it must be turned in on the very next day you are at school, even if your class does not meet that day (i.e. bring it to your teacher's room and turn it in).